

```

login as: root
root@hmc-laa-vm105 ~ # ps auxww
Last login: Wed Aug 19 01:33:56 2015 from 10.65.200.141
# NOTD for Thu May 21 10:12:04 EDT 2015
#-----
# HOST NAME: hmc-laa-vm105      IP: 10.65.200.141
# Kickstart Date:              Thu 21 May 2015 09:57:33 AM EDT
# OS Version:                  RHEL6.6 64-bit
#-----
# vCPU:                         2 cores
# CPU Chipset:                 Intel: IntelCore 3.0GHz 64-bit
#-----
# RAM:                         4 GB
# SWAP:                        4 GB
#-----
# Disk Drive Info
# /dev/sda:                     50.1 GB
#-----
# Note: This NOTD file gets updated automatically.
#       Changes to this file will be automatically overwritten!
# Please open a bug case if you have any problems with this machine
#-----
root@hmc-laa-vm105 ~ # cd /etc/puppet/puppet.conf
#-----
root@hmc-laa-vm105 ~ # cd /etc/puppet/puppet.conf
#-----
root@hmc-laa-vm105 ~ # cd /etc/puppet/puppet.conf
#-----
root@hmc-laa-vm105 ~ # /etc/init.d/puppet start
Starting puppet agent:                                     [ OK ]
root@hmc-laa-vm105 ~ # puppet agent --test
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Checking catalog for hmc-laa-vm105.1000.com
Info: Applying configuration version '140760219'
Notice: [main] /etc/manifests/site.pp/environments: created
Notice: Finished catalog run in 0.04 seconds
here is my Public IP address: 10.61.89.101
root@hmc-laa-vm105 ~ #
root@hmc-laa-vm105 ~ #
root@hmc-laa-vm105 ~ #

```

Figure 2: Result

Now create the manifest file of Puppet master touch, /*etc/puppet/manifests/site.pp*. After all this, start the Puppet master service with the following command:

```
/etc/init.d/puppetmaster start
```

Once Puppet is installed on the master, let's move on and customise it on the Puppet slave. Ruby must be installed on the slave now; so copy the repository package from the Puppet master server and enable the repository with the following command:

```
rpm -ivh puppetlabs-release-el-6.noarch.rpm
```

Now type `yum install puppet` to install the Puppet agent. After doing this, update the *puppet.conf* [agent] with the following line in quotes: “*server = <Puppet Master’s Hostname or FQDN>*”. Now start the Puppet agent service using the following command:

```
$ /etc/init.d/puppet start.
```

After successfully installing the Puppet on the master and client, we have to sign the SSL certificate on the master. The first time Puppet runs on an agent node, it will send a certificate-signing request to the Puppet master. Before the master is able to communicate and control the agent node, it must sign that particular agent node’s certificate. On the Puppet master, run the following command to list all unsigned certificate requests:

```
$ puppet cert list
```

```
+ “<Agent’s Hostname or FQDN>” (SHA256) 21:EF:ED:81:95:F8:ED:6D
:A5:1D:05:48:C9:01:5D:A6:FE:0C:FB:A4:89:FF:45:CF:26:53:1C:BB:1F
:6D:1E:CC
```

On the Puppet master, run the command given below to sign the agent’s certificate request:

```
$ puppet cert sign <Agent’s Hostname or FQDN>
```

The Puppet master can now communicate and control the agent to which the signed certificate belongs. Now that everything is up and running, let’s execute a simple test with Puppet. In the manifest file of the master, write some code to show the IP address.

Edit the main manifest file (*/etc/puppet/manifests/site.pp*) on the Puppet master with the following content, and then save it:

```

file {['/tmp/example-ip', # resource type file and filename
ensure => present, # make sure it exists
mode => 0644, # file permissions
content => "Here is my Public IP Address: ${ipaddress_eth0}.\n",
# note the ipaddress_eth0 fact
}

```

The Puppet agent periodically checks in with the Puppet master (typically every 30 minutes). It is also possible to initiate the check for a particular agent node manually, by running the following command (on the agent node):

```
$ puppet agent -test
```

Then run the following command to print the file:

```
$ cat /tmp/example-ip
```

The output looks like what’s shown below (with that node’s IP address):

```
Here is my Public IP Address: 10.61.89.101.
```

Thus we can configure the master and this configuration will be reflected in slave nodes. The benefits of this tool are:

- It frees up time to work on projects that deliver more business value.
- It ensures consistency, reliability and stability.
- It facilitates closer collaboration between sysadmins and developers, enabling more efficient delivery of clean code that delivers real business value. 

By: Ashish Sinha

The author is a software engineer based in Bengaluru. A software enthusiast by heart, he is passionate about using open source technology and sharing it with the world. He can be reached at ashi.sinha.87@gmail.com